

Does a Music Transformer Know What Key It Is In?

Causal Interventions on an Emergent Tonal World Model

Anonymous ICASSP 2027 submission

Abstract—Whether music generation models maintain an internal representation of musical key that generation *causally uses*—rather than chaining surface statistics—has not been tested. We test it under a pre-registered protocol on Transformers trained on symbolic music whose vocabulary holds absolute pitch and time only. Replacing the key component of the model’s hidden activations at a single middle layer makes the model continue in the injected key at $5.0\times$ the rate of rank- and norm-matched random edits (12/12 targets, Holm-corrected, replicated across seeds), within a musicality budget frozen before any edit ran. Training the same setup at four model sizes pulls apart what “world model” runs together: next-token accuracy is flat across a $170\times$ parameter range, yet the smallest model carries no decodable key state, and its edits move the key while wrecking the music 90% of the time. Both the probe and the intervention transfer to a public checkpoint trained on real music: its key state beats the pitch surface where our synthetic-trained twin’s does not, editing it redirects composition at $5.7\times$ the matched control (11/12 targets, 100% in budget), and the keys it resists are the keys real music rarely uses.

Index Terms—music generation, world models, interpretability, causal intervention, tonality

I. INTRODUCTION

Systems that compose music write convincingly tonal pieces [1], [2], [3]: they settle into a home key, build tension away from it, and return. How they do so is unexamined.

Sequence models can build internal models of the latent state behind their data. A model trained only to predict Othello moves tracks the board those moves imply, and editing that internal board changes which moves it treats as legal [4], [5]. Probing had earlier found rich game knowledge inside AlphaZero [6], and the same causal test now reaches chess, mazes, and RL environments [7], [8], [9]. Music has both halves of that toolkit: probes read musical attributes out of a model’s activations [10], and steering methods—adapting inference-time interventions from language models [11], [12]—change those attributes by acting on the activations themselves [13], [14].

However, neither half asks the question Othello answers. A signal can be decodable from a system’s state without the system consulting it when it acts [15]. And an intervention that moves an attribute establishes that a direction is causal,

not that the model maintains a *variable* along it: the test for a variable is whether one can set it to a different value—remove what is there and install something else, rather than push along an axis—and whether the model then composes as that value dictates. No one has run that test in music.

We run it, with musical key as the state variable—an unusually good choice. Theory and psychological measurement define it *outside* the model [16], so the state we probe is not an artifact of our labeling. The model must *compute* it by integrating pitch evidence over time, unlike metric position, which the time tokens hand over. And a key’s scale gives us Othello’s legal-move rate for free, so we can score success without a learned judge. We train Transformers on a vocabulary of absolute pitch and time alone, replace the key component of one layer’s activations with a donor key’s, and ask what they compose next.

We release the testbed—leak-free tokenizer, functional-harmony corpus with exact per-token key labels, frozen protocol—as a benchmark.

II. METHOD

A. Data, models, protocol

We need the true key at every note, and on real music the true key is itself an estimate, so we generate our training data. **D-SYN** (synthetic data) comes from a functional-harmony state machine: it walks the classical chord functions tonic→subdominant→dominant→tonic with the usual substitutions, voices each chord for four parts (soprano, alto, tenor, bass), and optionally adds a melody from the current scale. It writes 200k/10k/10k train/validation/test pieces of 278–508 tokens, each with 0–3 modulations (pivot, direct, or sequential; stratified by fifths distance). Because the machine knows which key it writes in, every token carries a true key label $\kappa(t)$ (tonic $\times$ mode, 24 classes). **These labels never reach the model**: they train and score the probe only. The vocabulary (124 types) holds only **BAR**, **POS** (position in the bar), **absolute-MIDI PITCH**, **DUR** (duration), and the structural symbols **PAD/BOS/EOS**; a unit test enforces the absence of key, chord, and degree tokens, so a model that knows the key must have computed it from the notes.

Models: **M-CTRL** (our controlled model) is a GPT-2 style Transformer decoder [17], [18] (8 layers, 8 heads, $d=512$, context 512, ~ 25 M parameters), trained under two regimes—with (**R-Aug**) and without (**R-NoAug**) transposing every piece into all 12 keys— $\times 3$ seeds, giving six models. **M-REF** (our reference model) shares the architecture but gets a different seed *and* a disjoint slice of the data. We never probe

or edit it: it only judges whether edited music still sounds like music, so that no model grades its own work. Every model clears a quality gate set in advance: validation next-token top-1 of 0.878–0.881 against the 0.434 of a most-frequent-token predictor. We froze all decision rules, thresholds and controls before any experiment ran; we log every deviation and report negative results.

B. Probing (Phase A)

A probe [19], [20] is a classifier that reads a quantity out of a network’s internal activations. We train linear probes to map the residual stream $h_\ell(t)$ —the hidden state that layer ℓ reads and updates—to the key $\kappa(t)$, splitting train and test by piece. Probe accuracy alone proves nothing, so three control families bound it. **C1** shuffles the key labels within each piece: a probe that still scores well is memorising, and we subtract whatever it scores—its *selectivity floor* [15]—from every margin. **C2** is an untrained model, which shows what architecture alone affords. **C3** is the one that matters: naming the key from the notes alone, without looking inside the model—logistic regression on a pitch-class histogram, and Krumhansl–Schmuckler (KS) template matching [21], [22], over windows $W \in \{8, 16, 32, 64\}$. C3 is the surface-statistics account made quantitative: only what the probe reads *beyond* C3 belongs to the model.

We pre-registered **H1** (*existence*): the key is decodable from internal activations, more accurately than from the input-surface baselines, and the gap widens where the key is ambiguous. **DR-H1**, the rule we froze for it: the probe supports H1 if and only if its selectivity-corrected F_1 beats the best C3 baseline in some layer, with a piece-level bias-corrected and accelerated (BCa) bootstrap [23] 95% confidence interval that excludes zero.

C. Intervention (Phase B)

We prompt the model with 8 bars in a stable key and let it continue. From bar 9 on, we edit one layer’s activations, following the activation-patching tradition [24], [25]:

$$h \leftarrow h - P_V h + P_V \mu_{\kappa^*}. \quad (1)$$

V is an orthonormal key subspace, $P_V = VV^\top$ projects onto it, and μ_{κ^*} is the average activation the model produces when writing in the donor key κ^* : we subtract h ’s key component and install the donor’s. The edit is *sustained*: we re-apply it at every position the model generates after bar 9—every **BAR**, **POS**, **PITCH** and **DUR** alike—because one edit alone would be re-estimated away by the next note. We therefore write the state more widely than we read it: the probe of Sec. III-A is trained and scored at sampled positions, uniformly over token types; only the public-model study (Sec. III-F) restricts reading to positions about to choose a pitch. **The prompt itself never changes.**

We build V three ways: **V-PROBE**, the probe’s row space (rank 24)—the directions the key can be *read* from; **V-MEAN**, spanned by the per-key mean activations; and **V-DAS**, optimised rather than read off by distributed alignment search [26], which trains it so that swapping the subspace between two runs swaps the key (ranks 8/24). Four controls bound the result. **K1** uses a random subspace matched in rank and norm: both bases are orthonormal ($\|V\|_F = \sqrt{24}$), and on real activations the two edits perturb the state by within 1% of each other ($\|\Delta\|$ 21.9 against 22.1, the control marginally the larger), so direction rather than magnitude separates them. **K2** is a sham edit that changes

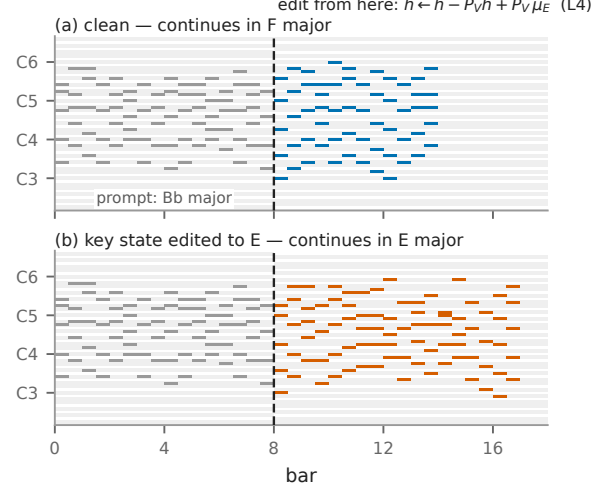


Fig. 1. The claim in one example (real model output, fixed sampling seed). The same 8-bar Bb-major prompt, continued (a) clean and (b) with the key component of layer-4 activations replaced by the E-major class mean from bar 9 on. **The input does not change**; the continuation’s estimated key follows the injected internal state. Shading marks the scale of the key in force.

nothing and must reproduce the clean run bit for bit (it did, 100/100). **K3** borrows another layer’s subspace. **K4** sets the ceiling by transposing the prompt itself—what full behavioural control looks like.

We generate 16-bar continuations for 12 major targets $\times 100$ prompts $\times 8$ layers, pairing every edited run with a clean twin so sampling noise cancels. Two measures score them: the *target-key rate* (TKR), the fraction of continuations KS estimation assigns to κ^* , reported strictly (the exact key; the supplementary adds a tolerant variant); and the *in-key rate* (IKR), the fraction of continuation pitches in a given key’s scale, which needs no estimator.

An edit that changes the key by wrecking the music is not a success, so we cap the damage. Before running a single edit, we measured how far M-REF’s perplexity rises across 7,989 *natural* modulations in validation data, and froze the 90th percentile of that rise as our musicality budget, $\delta_{\text{PPL}}=0.613$: an edit that costs more than a real modulation does not count. We call TKR scored under this rule *guarded*, and every headline number below is guarded.

H3 (*causality*, the central claim): a targeted edit of the key subspace moves the continuation’s key to the target, by a margin over a rank- and norm-matched random edit, and without leaving the musicality budget. **DR-H3**: at the best in-budget condition, $\text{TKR}(\text{edit})$ must exceed $\text{TKR}(\text{K1})$ with a Holm-corrected [27] Wilcoxon $p < .05$ in at least 8 of the 12 targets.

III. RESULTS

We report intervention numbers from the R-Aug seed-0 sweep. Probing holds for all six M-CTRL models, and the seed-1 sweep reproduces every Phase-B verdict, though it peaks at L2 rather than L4: the localization replicates, its depth does not.

A. The key is decodable beyond the input surface

The model computes the key rather than passing it along. At L4 the probe reaches macro- F_1 0.927, where the best surface baseline (C3: pitch-class histogram, $W=64$) reaches 0.790 and the selectivity floor is 0.032; the corrected margin is therefore $0.927 - 0.790 - 0.032 = +0.105$ (BCa 95% CI [0.091, 0.114]). That margin excludes zero in layers 1–7 of every trained model and fails only at L0, where it is significantly *negative* (-0.20 , CI $[-0.211, -0.182]$)—the embedding itself, which should not beat the surface: DR-H1 is supported 6/6. An untrained model scores 0.243, so the architecture alone does not supply the key.

One confound could explain that margin: the probe attends the full prefix (≤ 512 tokens) while C3 sees a W -token window, and the best C3 sat at the *edge* of the pre-registered sweep, still rising. So we extended the sweep to the full prefix and built two stronger variants (recency-decayed histograms; short+long concatenation). History does not close the gap: the window curve turns over at $W=96$ (0.801) and *falls* to 0.752 at $W=512$ —under modulations, stale evidence poisons a histogram that cannot weight it. Against the strongest surface so built (0.824), the margin is $+0.071$ (CI [0.056, 0.081]) and all twelve DR-H1 verdicts below are unchanged. One registered sub-claim does *not* survive: the probe’s edge is uniform across ambiguity strata ($+0.097$ in both), not wider where KS is uncertain—that contrast (0.936 vs. 0.795) was an artifact of the $W=64$ cap.

B. Fifths geometry; equivariant but not a group

The representation carries music-theoretic structure. We project the twelve major-tonic probe directions onto their first two principal components, and they arrange themselves around the circle of fifths (Fig. 2): every neighbour on the circle is a neighbour in the plot, where the chromatic ordering agrees not once. We found this after the fact, so we report it as exploratory.

Transposition behaves like a linear map that generalises: a Procrustes rotation R_k fit on half the pieces aligns held-out activations with 0.05–0.11 relative error. But the twelve maps do not form a cyclic group. **H2 (form)** predicted that they would—that transposing by k steps would be the same map as transposing by one step k times, $R_k \approx R_1^k$. We measure the shortfall as $\varepsilon_{\text{cyc}} = \text{mean}_k \|R_k - R_1^k\|_F / \|R_k\|_F$ over $k = 2 \dots 11$ ($k=1$ is identically zero and T_{12} is the identity), and get ≈ 0.98 : better than the ≈ 1.41 of random orthogonal maps, nowhere near the 0 a group would give. H2’s sub-claim **H2b** predicted that training on all twelve transpositions would tighten the family. It does not (0.983 against 0.981, three seeds each, one-sided $p = 0.83$), so DR-H2b is not supported.

C. Editing the key state changes the composed key

At the best in-budget condition (V-PROBE, L4), the edit beats its matched random control on all 12/12 targets (Holm-corrected $p < 10^{-4}$; rank-biserial $r = 0.64$ –1.00), against the 8/12 we had set in advance: **DR-H3 is supported** (Fig. 3). Raw pitch content says the same thing without the estimator: after editing, 93.7% of the notes belong to the injected key’s scale and 64.6% to the prompt’s. Success falls gently as the target moves away around the circle of fifths (0.395 at one step, 0.280 at the tritone), which is what moving a key should look like; random edits, once we exclude the target that asks for no change at all, land on the target no more often than chance.

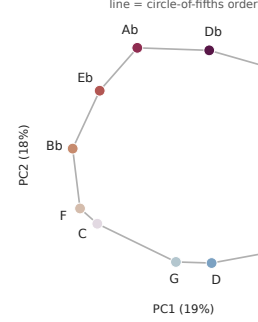


Fig. 2. The twelve major-tonic probe directions (L4, first two PCs) self-organize into a circle of fifths. No key, chord, or degree token exists in the vocabulary.

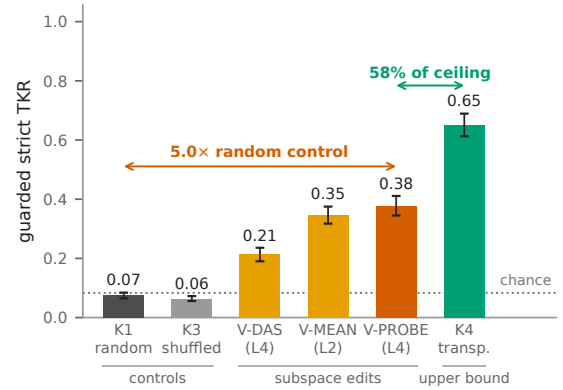


Fig. 3. Guarded strict TKR by condition: the continuation lands exactly on the injected key *and* stays inside the frozen musicality budget. Chance = 1/12. Bars: prompt-level bootstrap 95% CI.

The edit reaches 58% of the ceiling K4 sets. So the key is readable in a rank-24 subspace of a single layer, yet rewriting that subspace buys only part of the behaviour—an *actionability gap*. Two further findings. The *optimised* subspace (V-DAS, 0.213) does worse than the probe’s readout directions. And V-MEAN edits destroy the music at deep layers, where at most 5% of them stay in budget; without the guard those wrecked continuations would count as successes. Sec. III-E shows why that distinction decides the paper.

D. Readability is diffuse; causal efficacy is sharp

We pre-registered **H5 (localization)**: the layers at which an edit is causally effective are confined to a narrow band. Editing at each layer in turn and subtracting the matched control gives 0.03, 0.04, 0.19, 0.26, **0.30**, 0.26, 0.23, 0.23 for L0–L7: a single peak. The bootstrap CI for the best layer against the median layer, [0.038, 0.109], excludes zero, so DR-H5 is supported. Meanwhile the probe saturates across L2–L6 (macro- F_1 0.920–0.929, all within 0.01 of its maximum). The two curves in Fig. 4 are not the same shape, and that is the finding: the key is legible from most of the network, but a single-layer edit moves the output only near L4, and barely at all at L0–L1. *Where a state is legible is a weak guide to where it is used*—a caution for any work that locates a computation by probing accuracy alone.

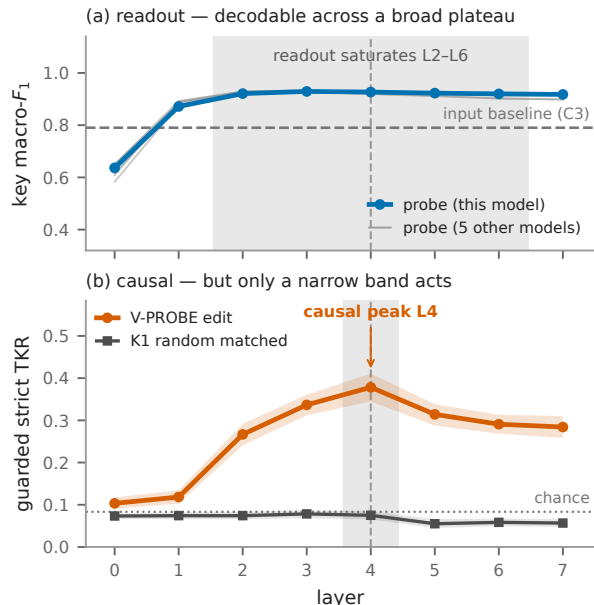


Fig. 4. (a) Probe macro- F_1 per layer, all six M-CTRL models (thin lines; R-Aug seed 0 highlighted): the key is legible across a broad plateau. (b) Causal effect: guarded TKR of the V-PROBE edit vs. the K1 matched random control (bands: prompt-level bootstrap 95% CI). Only a narrow band around L4 moves the output.

E. What a world model actually requires

We trained the same data with the same protocol at four sizes (L2/d128, L4/d256, L8/d512, L12/d768; 0.5M, 3.4M, 26M, 86M parameters). The results pull apart three things the term “world model” runs together (Table I, top). *Behaviour* is flat: next-token top-1 stays at 0.868–0.879 across a $170\times$ range. *Legibility* is not. At 0.5M the probe scores *below* the surface baseline and DR-H1 fails in both seeds—a model that composes about as well as the others and has no key state to speak of, the surface-statistics account in the flesh, which our pre-registered rule refuses to credit. (Every size verdict is unchanged against Sec. III-A’s strengthened baseline, but the 3.4M margin thins to $\sim +0.01$: the state appears just barely above the strongest surface.) *Manipulability* is a third property again. At 0.5M the edit moves the key *more* often than at 26M in raw terms (0.55 against 0.43), but only 10% of those edits stay in budget, against 80% at 26M: with two layers, “editing the key subspace” overwrites the model rather than moving a separable part of it. Report guarded TKR alone and this disappears; report raw TKR alone and a surface model passes for a world model. The guard tells the two apart. Nor does manipulability keep improving: it peaks near 26M and falls to 0.18 at 86M. We swept all twelve layers of the 86M model to check we had not merely picked a bad layer—its best, L6 at 0.206, is indistinguishable from the L3 we used, 0.202.

F. A public model trained on real music: it reads, and it uses

Everything above is our own model on our own data. We now leave both. We test on 300 Bach chorales—a long-standing benchmark for generative modelling [1], [28]—whose key at every note comes from published Roman-numeral

analyses.¹ Our 86M model’s key subspace transfers (F_1 0.534 against a 0.035 selectivity floor) but does *not* beat the surface: the corrected margin, $+0.034$ (CI $[-0.021, 0.103]$), does not exclude zero.

The Anticipatory Music Transformer [30] (Apache-2.0; Lakh MIDI [31], MetaMIDI [32], FMA [33], 450k commercial records) is, at its **small** size, architecturally identical to that model of ours—12 layers, $d=768$, and 85.1M non-embedding parameters against our 85.2M—and its vocabulary of (arrival-time, duration, note) triples leaks the key no more than ours does. We probe it on the same chorales, at the same positions, with the same controls and rule; the two C3 baselines land at 0.451 and 0.454, confirming the setups match. It reaches F_1 0.690, a corrected margin of $+0.196$ (CI $[0.132, 0.260]$): **DR-H1 supported**. Architecture, corpus, protocol and rule are fixed; *only the training data differs*, and it decides whether a key state is worth more than the notes themselves.

And that state is used. We ran the same intervention with three adaptations, each preserving a separation the protocol depends on. The subspace is the probe’s row space. The guard’s reference is a *different* public checkpoint (**medium** judging **small**), with the budget re-frozen at $\delta_{\text{PPL}}=0.849$ nats from 1,180 natural modulations in the chorales. We searched for the edit layer on 20 chorale prefixes and judged it on 60 it never saw; the sham edit again matched the clean run bit for bit (20/20). Unlike the main experiment, the budget was frozen after that layer search rather than before every edit—the search never consults it, and every number here postdates the freeze, but the guarantee is the weaker one. **DR-H3 is supported**: at L8 the edit lands the continuation on the injected key $5.7\times$ as often as the matched random control (0.365 against 0.064), significantly in 11 of 12 targets. Pitch content crosses over too (82.4% of notes in the injected key’s scale against 78.5% in the prompt’s). And where 87% of our own model’s edits stayed in budget, 100% of these do: a key state learned from real music is *more* cleanly separable, not less.

The real model shows two things a synthetic testbed cannot. Its causal peak (L8) is again not its probe peak (L10), so the dissociation in Sec. III-D is no artifact of synthetic data. And the model *resists*: the one target it cannot reach is $F\sharp$ major (TKR 0.000), and across all twelve targets success is ordered almost perfectly by how common that tonic is in the chorale corpus, counted over its note-weighted key labels (Spearman $+0.91$, $n=12$, one-sided permutation $p < 10^{-4}$; $+0.75$ counting each chorale once). A model trained on real music can be steered into the keys that music favours, and resists the rest—a prior our uniformly-keyed synthetic corpus could not have revealed. The chorale distribution is a proxy for how often tonal music uses each key, not a measurement of this model’s own training corpus (Lakh MIDI and MetaMIDI, not chorales), so we show that a prior exists, not where it came from. One practical caution: read the probe where the model is about to *choose* a pitch. Read it at the note, after the pitch is committed, and the same model’s profile decays with depth (F_1 0.545 at L0 to 0.093 at L10), the margin falls to $+0.042$ (CI $[-0.034, 0.092]$), and DR-H1 would be *unsupported*. One position choice, and the verdict inverts.

¹Scores: **bach-370-chorales** (CC BY-NC-SA); analyses: *When in Rome* [29] (CC BY-SA). Neither is redistributed.

TABLE I

WHAT A WORLD MODEL REQUIRES. **Top:** SAME DATA AND PROTOCOL, FOUR CAPACITIES — BEHAVIOUR IS FLAT WHILE LEGIBILITY AND MANIPULABILITY ARE NOT. **Bottom:** SAME ARCHITECTURE (12 LAYERS, $d=768$), SAME CHORALES, SAME PROTOCOL; ONLY THE TRAINING DATA DIFFERS. THE PUBLIC MODEL BOTH READS THE KEY AND USES IT.

Model	top-1	probe	DR-H1	guard
<i>capacity (D-SYN, our models)</i>				
0.5M	0.870	below C3	no	10%
3.4M	0.878	+0.05	yes	60%
26M	0.879	+0.10	yes	80%
86M	0.879	+0.10	yes	71%
<i>real music, matched architecture (Bach chorales)</i>				
ours, synthetic-trained	—	+0.034	no	—
public, real-trained [30]	—	+0.196	yes	100%

IV. DISCUSSION AND LIMITATIONS

Within this training condition, the evidence supports the world-model reading: a Transformer trained on notes alone maintains a mid-layer key variable, and generation consults it. Two observations say the model spreads that variable out rather than keeping it in one place: the edit reaches only 58% of the ceiling, and it fails at L0–L1. Othello-GPT, by contrast, is almost completely editable. The domain may explain it: an Othello board hides behind the moves, whereas every passing note re-announces the key, so the model can re-estimate it instead of storing it. A one-shot test confirms this directly: a value installed for a single bar shifts the pitch content durably (+0.12 over the matched control for 13 bars, no decay) yet splicing that bar’s *tokens* alone reproduces the entire effect — the state is re-estimated from the evidence, never latched. K3 adds a third: borrowing L0’s subspace does nothing at L4 (0.064, at or below K1), so the effect needs that layer’s own directions—but subspaces borrowed from L5–L7 and injected at L1–L3 reach 0.27–0.39, so the key subspace is largely *shared* across the middle stack rather than re-learned at each depth. The state is also spread across token *types*: restricting the write to PITCH positions is causally null (0.077, the K1 floor) while BAR/DUR positions alone carry 69% of the effect—and no restriction improves the guard rate, so the blanket write does not needlessly damage the music.

Bounds. *Mode:* we pre-registered major as the injected mode and minor as a secondary condition, and have not run the secondary condition, so every causal claim here is a claim about major targets. *V-DAS below V-PROBE:* the optimised subspace may be undertrained relative to its search space; we report it as observed and release its configuration. *Estimator:* the KS judge confuses keys a fifth apart; IKR triangulates around it here, and the supplementary brackets it with a tolerant TKR that accepts the relative, parallel, and fifth-neighbour keys. *Capacity:* the sweep varies depth and width together, so it locates a threshold without explaining it. *Selection:* the best condition is chosen on the data the test then uses, as pre-registered. *Real models:* the causal result in Sec. III-F holds for one public checkpoint at one size. *Persistence:* H4a, brought forward from its pre-registered later phase, bounds the claim: the key variable is causally consulted (H3) but not latched — persistence after release is carried entirely by the emitted notes. The rest of H4 (pivot

structure) remains future work. Supplementary material carries the ambiguity strata, the 12×24 specificity matrix, the fifths-distance curve, and per-seed variants of every figure. All our training data are synthetic; we use the evaluation corpora under their stated licenses.

V. CONCLUSION

We gave pre-registered causal evidence that a music Transformer tracks musical key as an internal state variable that generation consults, and measured how far a single-layer edit goes toward full behavioural control. Two controls say what such a state requires: model size buys not the key itself but a key that can be moved without breaking the music, and the training data—not the architecture—decides whether that state is worth more than the notes. We release the testbed (exact state labels, leak-free vocabulary, frozen decision rules) as a benchmark. Music offers state variables that theory defines and ears can verify; we ask next whether models also internalize how keys are *supposed* to change.

Reproducibility. Anonymized code, data generator, pre-registration, and an audio demo page (clean vs. edited continuations): <https://anonymous.4open.science/r/TBD>.

[TBD (run required): create before submission.]

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